

Chapter 25 — Molecular Genetics and Cancer Biology

- Molecular understanding of cancer follows from **Watson and Crick's structure of DNA (1953)**; the **first human genome sequence (early 2000s)** and expanding sequencing technology accelerated discovery of genetic and epigenetic determinants of cancer.
- **Carcinogenesis is multistep** and often takes many years; a normal cell must clear multiple hurdles.
- **At least eight attributes of the malignant cell**: (1) genetic instability and mutation; (2) autonomous growth; (3) insensitivity to antiproliferative signals; (4) resistance to apoptosis and other induced cell suicide; (5) unlimited division potential; (6) angiogenesis; (7) **local invasion — the feature that uniquely distinguishes malignant from benign**; (8) immune evasion.
- Cancer cells must also **tolerate the stresses** of their abnormal physiology and **increase energy metabolism**; **tumour-associated inflammation** can drive preneoplastic lesions to invasion; many acquire the lethal attribute of **metastasis**.

BASIC MOLECULAR GENETICS

Transcription

- **DNA → messenger RNA (mRNA) → protein**. **RNA polymerase II** synthesises pre-mRNA complementary to the template, containing **exons (coding)** and **introns (non-coding)**.
- Three nuclear modifications before translation: **5' capping**, **3' polyadenylation**, **splicing** (introns removed, exons joined).
- **Alternative splicing** lets one gene encode several proteins. **Androgen receptor (AR) splice variants lacking the ligand-binding domain** are active without ligand → may drive prostate cancer refractory to androgen ablation.

Protein Synthesis

- Translation occurs in the **cytoplasm on ribosomes**.
- **Frameshift vs substitution**: a single-base insertion or deletion shifts the frame and usually gives a non-functional protein; a single-base substitution may or may not change the amino acid, and may or may not alter function.
- **Transfer RNA (tRNA)**: 75–85-base adapter carrying one amino acid and a three-nucleotide **anticodon**. **Ribosomal RNA (rRNA)**: **~80% of cellular RNA**, part of the ribosome.
- **Stages: initiation** (often rate-limiting; 40S subunit binds mRNA; **start codon AUG → every protein begins with methionine**) → **elongation** (fastest; several ribosomes may read one mRNA at once) → **termination** at one of **three stop codons**.

Chromosome and Gene Structure

- Linear chromosomes are packaged by **histones into chromatin**; local chromatin structure governs expression and differs by cell type.
- **Euchromatin** (actively expressed DNA) vs **heterochromatin** (inactive, possibly structural).

SOMATIC AND GERMLINE TESTING

SOMATIC VS GERMLINE TESTING

	Somatic	Germline
Detects	Mutations acquired during life , confined to tumour	Inherited variants in every cell; transmissible
Sample	Tumour tissue (biopsy, resection)	Blood or saliva
Uses	Tumour landscape, targetable mutations → targeted therapy , heterogeneity, resistance mutations	Hereditary syndromes, risk assessment ; surveillance, preventive surgery, targeted therapy; family implications

INACTIVATION OF TUMOUR SUPPRESSOR GENES AND ACTIVATION OF ONCOGENES

Gene Mutation and Inactivation

- **Cancer is fundamentally a genetic disease.** DNA changes are frequent, but most are inconsequential or are detected and cause cell death.

CATEGORIES OF GENE ALTERATION

Category	Change	Consequence
Point mutation	Single nucleotide	Silent (no amino-acid change), missense (substitution), nonsense (premature stop codon)
Insertion/deletion ("indel")	Bases added or lost	Frameshift — large regions of protein altered
Duplication / deletion	Larger DNA segments	Loss of protein regions or whole genes
Chromosomal rearrangement	Inversions, translocations	Large deletions, duplication/amplification, gene fusions

BOX 25.1 GROSS CHROMOSOMAL ABNORMALITIES FREQUENTLY OBSERVED IN CANCER

Class	Abnormality
Numerical (aneuploidy)	Abnormal whole-set number (triploidy, tetraploidy); loss or gain of single chromosomes (monosomy, trisomy)
Structural — rearrangements	Inversions; translocations (balanced or unbalanced); chromothripsis
Structural — fusions	End-to-end → dicentric chromosome ; intrachromosomal → ring chromosome
Structural — dosage	Deletions (small segment to whole arm); duplications, amplifications; double minutes (often amplified sequence); isochromosome (one arm lost, replaced by duplicate of the other)
Complex	Combinations of the above

Tumour Suppressor Genes

- Negatively regulate growth; roles in **cell cycle, DNA repair, signalling**. Loss → dysregulated growth and division despite stress and damage → further mutation.
- **Both alleles usually must be lost**, by: (1) homozygous deletion; (2) loss of one allele + mutation of the other; (3) mutation of both; (4) loss of one allele + **epigenetic silencing (promoter methylation)** of the other. Classic examples: **RB1, TP53**.
- **Knudson's two-hit hypothesis (retinoblastoma)**: familial cases — often bilateral — inherit one non-functional allele, so a **single somatic hit** suffices.

- **Exceptions to two hits: dominant negative** — the mutant protein disables the wild-type partner where proteins act as multimers (e.g. **TP53**); **haploinsufficiency** — one lost allele lowers protein enough to raise carcinogen susceptibility (e.g. **CDKN1B** / **p27^{Kip1}**).

Oncogenes

- Mutated, activated forms of normal **proto-oncogenes** that drive proliferation. Activation by: (1) **activating mutation**; (2) **amplification** ($\uparrow$ copy number [CN] $\rightarrow$ $\uparrow$ transcript and protein; arises from replication errors, aneuploidy); (3) **chromosomal rearrangement**.
- Examples: **ERBB** mutation $\rightarrow$ constitutively active epidermal growth factor receptor (EGFR). **MYC** (early-response transcription factor): **amplified in prostate cancer**, linked to immortalisation of prostate epithelium; amplified/overexpressed in bladder cancer (more than half of papillary and invasive tumours). **MET** (hepatocyte growth factor receptor): overexpressed in RCC, more in high grade; **germline missense activation in hereditary papillary RCC**. **TPR2-ERG fusion** in a large proportion of prostate cancers.

TUMOUR SUPPRESSOR GENE VS ONCOGENE

	Tumour suppressor	Oncogene
Normal role	Brake on growth; repair	Accelerator (proto-oncogene)
Cancer change	Loss of function	Gain of function
Alleles needed	Usually both (exceptions: dominant negative, haploinsufficiency)	One
Mechanisms	Deletion, mutation, promoter methylation	Activating mutation, amplification, rearrangement/fusion
GU examples	RB1, TP53, CDKN1B, VHL, PTEN	MYC, MET, ERBB/EGFR, TPR2-ERG

KEY POINTS: TUMOUR SUPPRESSOR GENES AND ONCOGENES

- Somatic and germline testing identify alterations governing cancer risk and progression.
- Tumour suppressors restrain growth; lost by four routes; some escape two-hit rule (dominant negative, haploinsufficiency).
- Proto-oncogenes activated by mutation, amplification or rearrangement.

CELL CYCLE DEREGLATION

- Division is tightly balanced against cell loss; human cells integrate external **growth-stimulatory and growth-inhibitory signals** to choose division or quiescence. In cancer, activated oncogenes and lost suppressors tip the balance permanently towards proliferation.
- **G0** = reversible quiescence, the default. The cycle is ordered and unidirectional: **G1 $\rightarrow$ S (DNA synthesis) $\rightarrow$ G2 $\rightarrow$ M (mitosis)**, **about 24 hours**; each step requires completion of the previous.
- **Checkpoints** monitor DNA integrity $\rightarrow$ pause for repair $\rightarrow$ **apoptosis** if irreparable. Oncogenes and suppressors act largely by disabling checkpoints and apoptosis; loss of damage sensing $\rightarrow$ **genetic instability**, driving progression, resistance and metastasis.

Cyclin-Dependent Kinases and Cyclins

- **Cyclin-dependent kinases (CDKs)**: conserved **serine/threonine kinases** that phosphorylate phase-specific substrates.

- Activity depends on **cyclins (12 in mammals)**, whose levels rise and fall with the phase and are **rate-limiting**; after each phase the cyclin is **polyubiquitinated and degraded by the proteasome**, inactivating its CDK.
- Further control: **activating/inhibitory phosphorylation** of the CDK and **CDK inhibitors (CKIs)** — all commonly altered in cancer.

CYCLIN-CDK PARTNERS THROUGH THE CYCLE (FIG. 25.2)

Phase	Complex	Function
G1	Cyclin D-CDK4/6, cyclin E-CDK2	Cyclin D needs sustained mitogen signal (Ras/Raf/MEK); phosphorylate RB1
Early S	Cyclin A-CDK2 (A replaces E)	Activate S-phase genes
Late S	Cyclin A-CDK1	—
End of S → G2 → M	Cyclin B-CDK1	Drives mitosis; phosphorylates lamins → nuclear envelope breakdown

Cell Cycle Progression

- A cell proliferates when **growth-stimulatory signals (soluble growth factors) outweigh inhibitory ones (e.g. TGF- β)**. From G0 it enters G1 to build macromolecules and organelles; through early and mid-G1 it remains sensitive to external signals.

CELL-CYCLE CONTROL POINTS

Control point	Responds to	Mechanism	Outcome
G1/S checkpoint	Extracellular signals (mitogens, nutrients, oxygen) and DNA damage	No commitment without sustained signal; p21 (CDKN1A) and 14-3-3σ inactivate or sequester G1 CDKs	Arrest/repair; TP53 → apoptosis if unrepaired
S-phase arrest (not a formal checkpoint)	Genotoxic insult	ATM-dependent: (1) Cdc25A destroyed → CDK2 stays inactive; (2) ATM phosphorylates repair enzymes	Pause to repair — last chance before replication locks in mutations
G2/M checkpoint	DNA damage only (e.g. radiation)	ATM → CHEK1/CHEK2 phosphorylate Cdc25 → sequestered by 14-3-3 σ ; TP53 and BRCA1 contribute; TP53 induces p21, GADD45, 14-3-3 σ	G2/M arrest
Spindle assembly (mitotic) checkpoint	Any unattached chromosome ("not ready" signal)	Blocks the anaphase-promoting complex	Metaphase held until all attached

- **TP53 mediates both G1/S and G2/M arrest** — hence its central role in genomic integrity.

Mitosis

- **Sequence:** chromosomes condense (**condensins**); sister chromatids held by **cohesins** → **centrosomes** move to opposite poles and assemble spindle microtubules, which attach at **centromeres via kinetochores** → nuclear envelope and **lamins** dissolve (cyclin B-CDK1) → chromosomes align at midline (**metaphase**) → motors pull complements to poles (**anaphase**) → nuclei re-form, chromosomes decondense → **cytokinesis**.
- Mitotic abnormalities in cancer were noted by **von Hansemann** (late 1800s); **Boveri (1914)** proposed they cause tumours.

Mitotic Arrest: The Spindle Assembly Checkpoint

- Failed attachment → **non-disjunction**: one daughter trisomic, the other monosomic = **aneuploidy**, a **near-universal feature of solid tumours** and a common route to gene amplification and deletion; expression tracks CN, so aneuploidy may drive initiation and progression.
- Checkpoint-gene mutations (e.g. **hBUBR1**) are **rare**; **abnormal expression** of checkpoint proteins better explains instability — ↓ **MAD1/MAD2 in RCC**, ↓ **MAD2 in testicular cancer**.

Cyclin-Dependent Kinase Inhibitors and Checkpoint Inhibition

CDK INHIBITOR FAMILIES

	Cip/Kip	INK4 ("inhibitors of CDK4")
Members	p21 (CDKN1A), p27 (CDKN1B), p57 (CDKN1C)	p15 (INK4B), p16 (INK4A), p18 (INK4C), p19 (INK4D)
Range	Broad — many cyclin-CDK complexes throughout the cycle	Restricted to CDK4/6 → block RB1 phosphorylation → key G1/S regulators
Physiology	p21 ↑ after DNA damage ; p27 high in quiescent cells ; maintain G0 in terminally differentiated cells	p15 and p16 inactivated by mutation and promoter methylation in cancer, including GU tumours

Retinoblastoma (RB1) Tumour Suppressor

- RB1 controls the **late-G1 restriction point**, beyond which the cell is irreversibly committed to divide; continuous proliferation in cancer is largely from **loss of RB1 signalling**.
- Sequence**: hypophosphorylated RB1 binds **E2F** and recruits **histone deacetylases (HDACs)** → S-phase genes silent → during G1, **cyclin D-CDK4/6** and **cyclin E-CDK2** phosphorylate RB1 at eight sites → hyperphosphorylated RB1 releases E2F → E2F (at least seven members) dimerises with **DP1-3** → transcribes S-phase genes (DNA polymerase, S-phase cyclins) → after mitosis RB1 is dephosphorylated and represses E2F again.
- Inputs converging on RB1 (Fig. 25.3): **TGF-β** via p15/p27; **contact inhibition**; **DNA damage** via ATM/ATR → TP53 → p21; **oncogene stress** via p19^{ARF}; replicative senescence.
- Ways cancers disable RB1**: mutation or deletion; promoter hypermethylation; overexpression/amplification of cyclin D or E2F; loss (mutation, deletion, methylation) of CDKIs. **The RB1 pathway (cyclins D/E, CDK4, E2F, p16, p27) is probably altered in almost all human cancers.**

TP53 Tumour Suppressor

- TP53 links checkpoints to genome integrity: DNA damage (via **ATM/ATR** → **CHK1/CHK2**) or oncogene activation (via **ARF**, which blocks MDM2) → phosphorylated, active TP53 → **cell-cycle arrest, DNA repair, senescence or apoptosis**; **MDM2** is its negative regulator.
- TP53 is the most commonly mutated gene in cancer** and in a pan-cancer screen of metastatic disease; prominent in GU malignancies.

Concept — *The cycle is a one-way ratchet driven by cyclin-CDK pulses and braked by CDKIs; RB1 guards entry (G1/S) and TP53 guards integrity (G1/S and G2/M). Cancer needs to break both, which is why these two pathways are lost in almost every tumour.*

KEY POINTS: CELL CYCLE DEREGLATION

- Phase-specific CDK phosphorylation orchestrates G1 → S → G2 → M.
- G1 → S depends on RB1 phosphorylation**; RB1 mutations common in urological cancer.

- Main control points G1/S and G2/M use CDKIs in response to DNA damage, contact, cytokines, hypoxia.
- TP53: arrest and repair, or apoptosis if damage irreparable.

DNA DAMAGE AND REPAIR

- DNA polymerase plus error correction: **about 3 incorrect bases per division in a genome of more than 3 billion.**
- Cancer-causing change still arises via **mutation, CN alteration and epigenetic change** (methylation, chromatin condensation, promoter access), from **endogenous** (mitochondrial respiratory by-products) and **exogenous** (chemicals, radiation) agents.
- **Prevention:** free-radical scavengers (**α -tocopherol, vitamin C, carotenoids, bilirubin, urate**) and detoxifying enzymes (**superoxide dismutases, glutathione peroxidases, glutathione transferases**).

DNA Repair Mechanisms

- About **10 000 oxidative lesions per cell per day**, plus spontaneous depurination, depyrimidination and deamination.
- **DNA damage response (DDR)** = replication proofreading + specific repair systems, tightly integrated with the cycle: first step is **arrest**; **ATM and ATR** activate TP53, CHK1 and others.
- **Loeb's mutator hypothesis:** the spontaneous mutation rate is too low to explain the number of mutations in most cancers → early loss of replication-fidelity genes produces a **hypermutable phenotype**, which is harder to treat because therapy rapidly selects resistant clones.
- Recessive cancer-predisposition syndromes caused by repair defects — **xeroderma pigmentosum, Bloom, Werner, ataxia telangiectasia, hereditary breast and colon cancer** — implicate hypermutability in causation.
- **Three systems repair damage to bases themselves:** NER, BER, MMR.

DNA REPAIR PATHWAYS (FIG. 25.5)

Pathway	Lesion	Causes	Key proteins	Cancer link
Nucleotide excision repair (NER)	Bulky, helix-distorting lesions, cross-links	Ultraviolet , chemicals	XPC, XPD, XPE/DDB1, ERCC1/XPF; CSA/CSB (transcription-coupled)	Xeroderma pigmentosum; ERCC2 in bladder cancer
Base excision repair (BER)	Single-base damage, single-strand breaks; no major distortion	Deamination, radiation, reactive oxygen species (ROS) , alkylating agents, replication errors	MUTYH, OGG1 glycosylases (A/G, A/C mismatches); PARP1/2; XRCC1	MUTYH → colorectal cancer; OGG1 variants → RCC
Mismatch repair (MMR)	Mispaired bases; insertion/deletion loops (1 to ≥10 bases) from slippage at repeats	Polymerase incorporation errors	MSH2/MSH6, MLH1 /PMS2	Defect → ~100-fold ↑ mutation rate ; MLH1 loss in colon, prostate, bladder cancer
Homologous recombination (HR)	Double-strand break (DSB)	Ionising radiation, free radicals, chemicals, replication of nicked DNA	ATM, MRN complex, RPA, RAD51, BRCA1, BRCA2, BARD1	Familial breast/ovarian cancer; prostate cancer

Pathway	Lesion	Causes	Key proteins	Cancer link
Non-homologous end joining (NHEJ)	DSB	As HR	Ku70/Ku80 (XRCC6/XRCC5), DNA-PKcs (PRKDC), Artemis, XRCC4-ligase IV	Rearrangements, translocations

Double-Strand Break Repair

- **DSBs are potentially the most disruptive lesion**; unrepaired → apoptosis — the basis of **radiotherapy and many chemotherapeutics**, so cells that abrogate this response resist therapy. Misrepair → mutations, deletions and the **structural rearrangements nearly ubiquitous in solid tumours**.
- **DSB response sequence**: break recognised → **ATM autophosphorylates/activates** → phosphorylates TP53, CHEK1, CHEK2 (arrest) → phosphorylates histone variant **H2AX** → **γH2AX**, which spreads up to **30 Mbp** either side and recruits DDR factors into **repair foci**.

HOMOLOGOUS RECOMBINATION VS NON-HOMOLOGOUS END JOINING

	HR	NHEJ
Template	Undamaged sister chromatid	None — ends ligated directly
Fidelity	Precise	Error-prone ("most extreme"); nucleotides lost; wrong ends joined → translocations
Cycle phase	Only after S phase, in G2, before mitosis	Any
Sequence	RAD51 with BRCA1, BRCA2, BARD1 copies the sister template	Ku binds ends → activates DNA-PKcs → phosphorylates Ku and H2AX → Ku helicase unwinds → ligase IV/XRCC4 seals

- **Ploidy changes** (tetraploidy, octaploidy) arise from non-disjunction, **endoreduplication** (doubling without division), cytokinesis failure and cell fusion.

Concept — *Each repair pathway is matched to a lesion (bulky → NER, subtle base → BER, mismatch → MMR, DSB → HR or NHEJ), and each inherited defect produces a cancer syndrome. The same logic makes repair defects therapeutic targets: HR loss → PARP inhibitor, MMR loss → immunotherapy.*

KEY POINTS: DNA DAMAGE AND REPAIR

- DDR = damage recognition and repair; damage rarely causes cancer because repair is redundant.
- NER — ultraviolet/chemicals; BER — deamination, oxidation, alkylation; MMR — polymerase mispairs; DSB repair — ionising radiation, free radicals.
- Inherited repair defects → marked cancer susceptibility.

APOPTOSIS AND CELL DEATH

APOPTOSIS VS NECROSIS

	Apoptosis	Necrosis
Nature	Programmed, energy-requiring, orderly	Non-programmed, acute injury (e.g. trauma)
Morphology	Cytoplasm shrinks, membrane blebs , chromatin condenses → apoptotic bodies engulfed by neighbours/macrophages	Cell swells and bursts , releasing contents
Inflammation	Little	Potent
Examples	Developmental remodelling; deletion of self-reactive T cells	Physical trauma

- Irreparable DNA damage → apoptosis → replacement from undamaged cells; hence **defective apoptosis promotes cancer**.

- Apoptosis also underpins **immune killing of tumour cells** (surveillance and immunotherapy) and the response to **chemotherapy and radiotherapy** — defects confer resistance.

Regulators of Apoptosis

- Balance of positive and negative signals; **feed-forward loops make it irreversible** once started. Signals converge on **caspases** (cysteine proteases with aspartate specificity; at least 13).
- **Initiator caspases 8, 9, 10** → **executioner caspases 3, 6, 7**; made as inactive **procaspases**, activated by cleavage in an amplifying cascade.
- **Executioner targets**: anti-apoptotic Bcl-2 and Bcl-XL (cleavage releases pro-apoptotic fragments) → repair/replication proteins (DNA-PKcs, replication factor C) → nuclear structure (lamins, NuMA, SAF-A) → cytoskeleton (keratin, actin) → adhesion (β -catenin, focal adhesion kinase).

The Intrinsic Apoptotic Pathway / The Extrinsic Apoptotic Pathway

INTRINSIC VS EXTRINSIC APOPTOSIS (FIG. 25.6)

	Intrinsic (mitochondrial)	Extrinsic (death receptor)
Trigger	Internal stress : unrepaired DNA damage; loss of survival signals (cell-cell or cell-matrix contact, hormone or growth-factor withdrawal)	Extracellular ligand on a death receptor — TNFR1, Fas (CD95) (TNF superfamily)
Sequence	Bcl-2 balance shifts pro-apoptotic → outer-membrane permeability $\uparrow$ → cytochrome c released → binds Apaf-1 = apoptosome → caspase-9	FasL trimer binds three CD95 → death domains recruit FADD (also TRADD, RIP) → death-effector domain binds procaspase-8 (FLICE) → autoactivation → caspase-8 (also 10)
Brakes	Anti-apoptotic Bcl-2 family; IAPs inhibit caspases or their activation	FLIP binds FADD, blocking procaspase-8
TP53	TP53-dependent (transactivates Bax)	TP53-independent → attractive target when TP53 is mutated

BCL-2 FAMILY

Pro-apoptotic ($\geq$ a dozen)	Anti-apoptotic ($\geq$ half a dozen)
Bax, Bak, Bok, Bik, Bad, Bid, Bim	Bcl-2, Bcl-XL, Bcl-W, Mcl-1
Bax (best studied): transactivated by TP53 ; dimerises in the mitochondrial membrane to form a cytochrome c channel	Bcl-2 (best studied): binds Bax , blocking permeability
Bid : cleaved by caspase-8 → activates Bax, inhibits Bcl-2 — the link from extrinsic to intrinsic	All share BH1-BH4 domains that allow heterodimerisation

- Many therapies (chemotherapy, external-beam radiotherapy) need **TP53** to induce apoptosis, yet **TP53 is mutated in more than half of malignancies**.

Concept — Both routes converge on the caspase cascade; the **intrinsic pathway** reads internal damage through Bcl-2 and mitochondria and needs TP53, the **extrinsic pathway** reads external death ligands and does not — which is why death receptors matter when TP53 is lost.

KEY POINTS: APOPTOSIS AND CELL DEATH

- Apoptosis is regulated, energy-requiring and irreversible.
- Intrinsic: Bcl-2 family governs cytochrome c release; extrinsic: death receptors activate caspases.
- TP53 is central to apoptosis but mutated in over half of cancers.

Senescence, Telomeres, and Autophagy: Protection Against Cancer

- **Senescence and autophagy** (linked to apoptosis) handle oxidative damage and stress of ageing and limit proliferation and immortalisation; once cancer exists they also mediate **resistance to radiotherapy and cytotoxics**.

Replicative Senescence

- First seen in cultured cells after replicative exhaustion (**Hayflick**); linked to **loss of DNA methylation and telomere shortening**. DNA damage, oncogenic signalling and oxidative stress cause **induced/accelerated senescence**.
- **Senescent cells are viable, metabolically active but growth-arrested** (G1 or G2/M) via **p16^{Ink4a}, p21^{Waf1/Cip1}, p27^{Kip1}**; they **persist**, unlike cells dying by apoptosis, autophagy or mitotic catastrophe.
- **Senescence-associated secretory phenotype (SASP)**: cytokines, chemokines, matrix-remodelling enzymes — immunostimulatory but potentially pro-tumour.
- Linked to ageing, **prostate cancer and benign prostatic hyperplasia (BPH)**. Morphology: flattened, enlarged, prominent nucleus, granular cytoplasm; stains for **senescence-associated β -galactosidase (SA- β -gal)**.
- Often bypassed in cancer. **p53-dependent** route (ATM, Chk2, ATR → **p21**, key effector; forced p21 induces senescence even without p53; **p63, p73** also involved; **ARF** stabilises p53 by blocking **MDM2**) vs p53-independent routes; transformed cells lacking p53 and Rb retain some capacity to senesce.
- **Therapy-induced senescence (TIS)**: **ADT induces senescence** in a subset of prostate cancers, and SASP may explain ADT-associated inflammation; **enzalutamide** can induce senescence via ↓ replication-licensing factor **CDC6**, partly through **GATA2**. Senescent **stromal fibroblasts** may promote cancer; removing senescent cells may improve chemotherapy → senescence is a **double-edged sword**.

Telomeres and Telomerase

- **Telomeres**: repetitive chromosome-end sequences protecting against degradation and recombination; **lose ~100 bp per division** (lagging-strand RNA priming; polymerases cannot copy the ends) → the "**mitotic clock**".
- **Telomere sequence to cancer (Fig. 25.7)**: proliferation, oxidative stress, DNA damage → **moderate shortening** → **DDR** → **senescence or apoptosis** (barrier 1) → if checkpoint inactive, further erosion → **severe shortening** → **chromosomal instability (CIN)** generating oncogenic mutations but also lethal damage (barrier 2) → **telomerase activation** → **stabilised telomeres and immortalisation** → **cancer**; without it, death from CIN.
- Most cancers have **abnormally short telomeres**; telomere-specific fluorescence in situ hybridisation (FISH) shows prostate cancer cells shorter than adjacent normal epithelium. **Premalignant bladder and prostate lesions** already short → early, causal role.
- **Short telomeres in primary prostate cancer predict recurrence** independent of age, Gleason sum and nodal status; short, variable telomeres in stroma → higher prostate cancer death risk.
- **Telomerase**: specialised **reverse transcriptase** adding telomeric repeats; restricted in somatic tissue to rapidly proliferating tissues (**intestinal epithelium, endometrium, testis, activated lymphocytes**). Forced expression → immortalisation. Supplies **two functions: quelling CIN and unlimited replication**. **85–90% of human tumours express high telomerase**.
- **Alternative lengthening of telomeres (ALT)**: telomerase-independent recombination, associated with **ATRX or DAXX** inactivation.
- **Therapy: imetelstat** (telomerase inhibitor) in myelodysplastic syndrome; small molecules, peptides and hTERT-based immunotherapies (**UV1, GX301, INO-9012**) in early prostate, renal and bladder trials.

Autophagy

- Conserved intracellular self-defence: organelles and cytoplasm sequestered in **autophagic vesicles** → **fused with lysosomes** → **degraded**. Discovered in yeast (**Nobel Prize, Ohsumi**).
- Low basal level; **upregulated by starvation, growth-factor withdrawal**, aggregated proteins, damaged organelles, intracellular pathogens. **Above the "autophagy threshold"** → **death** by necrosis, mitotic catastrophe or apoptosis. **Autophagy-dependent cell death** is a defined form of regulated death (Nomenclature Committee on Cell Death).
- **Signalling**: ROS, growth factors, metabolic signals → **mTOR and AMPK** → **ULK1/ULK2** → phosphorylates **Beclin-1** → autophagy-related (**ATG**) genes (ATG5, ATG12; about 30 known).
- **Context-dependent role**: **BECN1** loss raises cancer in mice and is frequent in human breast and prostate cancer (**tumour suppressor**); in early (e.g. KRAS-driven) tumours autophagy limits instability, inflammation and damage; but in hypoxic, nutrient-poor tumour cores it **supports survival**; in BRAF models it suppresses benign tumours yet its inhibition accelerates advanced cancers.
- Therapy-induced autophagy is usually **cytoprotective**. KRAS-inhibited tumours become autophagy-dependent → **MEK/ERK + autophagy inhibitors**. **Chloroquine/hydroxychloroquine** (lysosomotropic, raise pH) may synergise with chemotherapy; after **enzalutamide**, clomipramine or chloroquine increased apoptosis in CRPC cells. **No impact yet on urological cancer management**.

SENESCENCE VS AUTOPHAGY AS ANTICANCER DEFENCES

	Senescence	Autophagy
Cell fate	Viable, permanently arrested	Survival under stress; death above threshold
Marker/driver	SA- β -gal; p16, p21, p27; telomere shortening	Beclin-1, ULK1/2, ATG genes
Therapeutic dilemma	Induce (TIS) vs remove (SASP pro-tumour)	Induce vs inhibit (context-dependent)

Concept — *Senescence, telomere attrition and autophagy are barriers that stop damaged cells proliferating, but each can be co-opted by an established tumour to survive therapy — hence the recurring "induce or inhibit?" dilemma.*

KEY POINTS: SENESCENCE, TELOMERES AND AUTOPHAGY

- Replicative senescence = growth arrest from methylation loss and/or telomere shortening.
- Telomeres = mitotic clock; shortening is a barrier to unrestrained growth.
- Autophagy clears aggregates, organelles, pathogens; extreme stress beyond the threshold kills.

Cancer Epigenetics

- Disrupted epigenetic programming is a **hallmark of cancer**: **DNA methylation, histone modification and non-coding RNAs (ncRNAs)** alter chromatin accessibility → silence repair and cell-cycle (suppressor) genes and permit growth-promoting genes.
- **Reversible** → drug target: monotherapy success in **haematological** tumours; combinations with hormonal, chemo- or immunotherapy promising in solid tumours.

DNA Methylation

- Methyl group on **carbon 5 of cytosine** → **5-methylcytosine**, mainly at **CpG** dinucleotides; essential for embryonic development. CpG is **depleted genome-wide** because methylcytosine deaminates to thymine.
- **CpG islands** lie upstream of **~40% of gene promoters**. **Hypermethylation** → **silencing** (altered chromatin, blocked transcription-factor binding, excluded machinery); **hypomethylation** → **expression**.

METHYLATION CHANGES IN CANCER (FIG. 25.8)

	Promoter CpG island hypermethylation	Global / repeat hypomethylation
Effect	Silences tumour suppressors	Genomic instability (chromatin change, recombination between repeats, reactivated transposons); oncogene/imprinted gene re-expression
GU examples	GSTP1 — commonest somatic alteration in prostate cancer (>90%); p16/INK4A (27–60% of urothelial carcinoma); MGMT, MLH1; RASSF1A in up to 97% of bladder tumours	LINE-1 hypomethylation in bladder cancer (↓ DNMT1, ↑ DNMT3A/3B); IGF2 re-expression in prostate cancer via CTCF binding (loss of imprinting)

- **DNMT1** maintains methylation; **DNMT3A/3B** methylate de novo.
- **Mutagenic effect:** 5-methylcytosine deaminates spontaneously to thymine → **C→T transitions**. Tumours show a further **20–50% reduction** in methylcytosine at individual CpGs.
- Methylation changes are **among the earliest and most frequent events**, may precede morphological change, and in lethal metastatic prostate cancer are **shared across all metastases within a patient** → driver events.
- **Non-CpG methylation** (cytosine before A, T or C) is a newer regulatory mark.

Histone Posttranslational Modifications

- **Nucleosome:** octamer of two each of **H2A, H2B, H3, H4** wrapped by ~**146 bp** of DNA. N-terminal tails are reversibly **acetylated, methylated, phosphorylated, ubiquitinated** — altering chromatin and recruiting regulators; hundreds of combinations suggest a "**histone code**", kept in balance by writer/eraser enzymes and "reader" proteins.

ACTIVATING VS REPRESSIVE HISTONE MARKS

Activating	Repressive
H3K4me3 , H3K36me2; acetylation of H3K9, H3K14 and H4K5, K8, K12, K16 (opens chromatin)	H3K9me3, H3K27me3 (EZH2 target)

- **Histone acetyltransferases (HATs):** e.g. **p300** increases **AR** expression/activation → **AR** signalling without **androgen**; others raise IGFBP-2 and p21 to slow the cycle. **Loss of H4K16ac and H3ac** is a **common cancer hallmark**.
- **EZH2** (histone methyltransferase, **H3K27**): ↑ in hormone-resistant metastatic prostate cancer; **strong staining in primary tumour predicts PSA recurrence**; with HDAC1 silences growth inhibitor **hDAB2IP**.
- Global histone acetylation/methylation patterns stratify recurrence risk; **H3K4 methylation and H3K18 acetylation** independently predict recurrence. Hyperacetylation accompanies nuclear-hormone-receptor activation; **Jumonji C** demethylates H3K9me3 at AR targets.
- **HDAC inhibitors** (suberoylanilide hydroxamic acid [SAHA], phenylbutyrate): pro-apoptotic, growth-inhibitory, radiosensitising, antiangiogenic in vitro; phase I in solid tumours including prostate showed acetylated H3/H4 accumulation.
- Approved first-generation agents: **DNMT inhibitor for myelodysplastic syndrome**, **HDACi for cutaneous T-cell lymphoma** — non-selective, limited in solid tumours, toxic. Newer: inhibitors of **BET, EZH2, DOT1L** for tumours with mutations/translocations in these genes; sequential or sub-maximal dosing may help combinations.

Noncoding RNA

- ~**98% of the eukaryotic genome** is transcribed as ncRNA; <**2% encodes protein**.

NON-CODING RNA CLASSES

Class	Size	Action	GU examples
Long ncRNA (lncRNA)	>200 nt; no open reading frame; ~19 000 annotated	Signal, decoy, guide, modulate miRNAs; oncogenic or suppressive; 3D structure binds DNA, mRNA, protein	HOTAIR blocks AR binding to E3 ligase → castration resistance; MALAT1 ↑ in renal and prostate cancer
MicroRNA (miRNA)	18–25 nt ; ~500 human	Fold into hairpins; bind 3' untranslated region → silence many mRNAs; "oncomirs" or suppressors	Biomarkers in prostate, kidney; miR-371a-3p in testis
Small interfering RNA (siRNA)	Short, from long double-stranded RNA	Silences one specific mRNA	—
Piwi-interacting RNA (piRNA)	24–31 nt, single-stranded	With PIWI proteins : DNA and RNA methylation, histone modification, mRNA degradation; controls transposons	Emerging biomarkers

- lncRNAs dysregulate **STAT3, NF-κB, Wnt, PI3K/Akt, PTEN**. Methylation, histone marks and miRNAs are detectable in **blood and urine** → **liquid biopsy**.

Concept — *Epigenetic change alters how much of a gene is read, not what it says: promoter hypermethylation mimics a second hit on a suppressor, global hypomethylation destabilises the genome, and all of it is reversible — the reason it is both an early biomarker and a drug target.*

KEY POINTS: EPIGENETICS

- 5-methylcytosine deaminates → C→T transitions.
- Promoter hypermethylation silences suppressors; hypomethylation → oncogene expression, instability.
- Histone modifications change chromatin conformation and expression.
- siRNA: one target; miRNA: many. ncRNAs drive oncogenesis and resistance; detectable in liquid biopsy.

Angiogenesis, Hypoxia, and HIF-1

- **Angiogenesis is essential for proliferation and metastasis** and a major target in **renal** and other cancers; it reflects a balance of pro- and anti-angiogenic regulators from stroma, leucocytes and tumour cells.

Hypoxia-Inducible Factor (HIF) and Von Hippel-Lindau (VHL) Pathway**HIF REGULATION: NORMOXIA VS HYPOXIA (FIG. 25.10)**

Normoxia	Hypoxia (or VHL loss)
HIF-α (HIF-1α or HIF-2α) prolyl-hydroxylated → bound by VHL (E3 ubiquitin ligase with elongins, CUL2, RBX1) → ubiquitinated → proteasomal degradation	HIF-α-VHL binding falls → HIF-α accumulates, dimerises with HIF-1β (ARNT) → nucleus → binds hypoxia-response elements → VEGF, EphA1, angiopoietin and other genes that raise oxygen delivery

- HIF-1 is the **general oxygen sensor**; it also drives **vasculogenic mimicry** (Claudin-4, vimentin, Twist1): **tumour cells, not endothelium, line tortuous, hyperpermeable channels**.
- **VEGF upregulation correlates with microvessel density, advanced stage, progression and poor prognosis in renal tumours**.

Hypoxia and Metabolism

- Tumour oxygen tension is low; via HIF-1, cells shift to **↑ glucose transport, glycolysis to pyruvate and lactate, ↓ mitochondrial metabolism and mass**.

- **CA9, MCT4, NHE1** keep **intracellular pH alkaline (proliferation)** and **extracellular pH acidic (invasion)**; HIF-1 α represses angiogenesis inhibitors such as **thrombospondin-2**.
- **Warburg effect** (Nobel Prize 1931): aerobic glycolysis — best understood in **clear cell RCC**, where **VHL loss by deletion or methylation** → **constitutive HIF-1 α and HIF-2 α** . VHL is inactivated in **roughly 60% of RCC**, by somatic mutation (~50%) or promoter methylation (~10%) (clear cell-specific figures: see below).
- Hypoxia induces **N- and E-cadherin, vimentin, MMP2, MMP9** → invasion and metastasis.
- **Immune effect**: HIF-1 α , VEGF, IL-6, IL-10, PD-L1 and TGF- β enhance **MDSC** suppression, recruit **tumour-associated macrophages** and block **DC maturation** → modulating this environment is central to modern renal cancer therapy.

Concept — VHL loss is permanent pseudohypoxia: *the cell behaves as if starved of oxygen, and angiogenesis, Warburg metabolism, invasion and immunosuppression all follow from that single signal — which is why clear cell RCC responds to VEGF, HIF-2 α and immune blockade.*

Cell Plasticity and Pluripotency

- **Stem cells**: self-renewal, generation of progeny, specialised niches, ability to form all cell types of an organ; share with cancer cells **expanded replicative potential, telomere elongation, damage resistance, undifferentiated state** — debate whether cancers co-opt stem properties or arise from stem cells.
- **Lineage plasticity**: epigenetic and phenotypic change, **usually after extreme pharmacological targeting of a driver**; cells trans-differentiate to another lineage with new drivers and biomarkers.

CANCER STEM CELL VS STOCHASTIC (CLONAL EVOLUTION) MODEL (FIG. 25.11)

Cancer stem cell	Stochastic / clonal evolution
Hierarchy : a small population of resilient, non-proliferative cells gives rise to proliferative, therapy-sensitive cells — explains strong response then recurrence	No hierarchy : individual cells with distinct phenotypes, any of which may survive therapy and regenerate the tumour

NEXT-GENERATION SEQUENCING TECHNIQUES

- **Next-generation sequencing (NGS)**: rapid, cheap, high-throughput **massively parallel** sequencing of millions of fragments; profiles genomes, transcriptomes and epigenomes.

Next-Generation Sequencing Workflow

- **Sequence (Fig. 25.12)**: extract and **fragment** DNA/RNA → ligate **adapters with barcodes** → **amplify into a library** → **immobilise** on a solid surface → add **fluorescently labelled nucleotides**, record emissions → millions of short **reads** → **align** to reference or assemble de novo → bioinformatic analysis (variants, expression, epigenetic marks).

NGS TECHNIQUES

Technique	Measures	Key feature
Whole-exome sequencing (WES) (DNA Sequencing)	Protein-coding exome	Cheaper, focused; somatic point mutations, indels, CN; drivers vs passengers; new genes; germline predisposition
RNA sequencing (RNA-seq)	Whole transcriptome	Expression of all genes; differential expression; alternative splicing, fusion transcripts ; miRNA, lncRNA. WES = mutational landscape; RNA-seq = expression
DNA methylation sequencing	Methylated cytosines	Bisulfite conversion : unmethylated C → uracil; methylated C unchanged; compare with untreated DNA
Single-cell RNA-seq (scRNA-seq)	Transcriptome per cell	Cells sorted and barcoded before library , so each read traces to one cell; cell types, states, heterogeneity

Technique	Measures	Key feature
ATAC-seq (assay for transposase-accessible chromatin)	Open chromatin	Hyperactive transposase inserts adapters into accessible regions; enhancers, promoters
ChIP-seq (chromatin immunoprecipitation)	Protein-DNA binding	Cross-link, antibody pull-down of protein-DNA, sequence the DNA; transcription-factor sites, histone marks

KEY POINTS: NEXT-GENERATION TECHNIQUES

- NGS profiles genome, transcriptome and epigenome.
- WES sequences the exome; RNA-seq analyses the transcriptome.

ADVANCES IN THE UNDERSTANDING OF MOLECULAR GENETICS IN GENITOURINARY TUMOURS

Prostate Cancer Somatic Alterations

- Clinical heterogeneity (indolent to lethal) mirrors molecular heterogeneity; **The Cancer Genome Atlas (TCGA)** defined key drivers of localised disease.

KEY SOMATIC ALTERATIONS IN PROSTATE CANCER

Gene	Alteration	Significance
TMPRSS2-ERG	Fusion: androgen-responsive TMPRSS2 promoter/regulatory region drives ERG (ETS factor) body	Commonest fusion, ~half of prostate cancers ; other ETS partners ETV1, ETV4, ETV5 less often
FOXA1	Pioneer factor; mutation, structural variant (SV), amplification	Class 1 (wing-2 forkhead mutations): early, activate AR programme; class 2 (C-terminal truncation, ↑ DNA affinity): metastatic; class 3 (rearrangements repositioning FOXMIND enhancer → overexpression): metastatic
MYC	8q whole-arm gain, focal amplification, enhancers	8q24 = germline susceptibility region
TP53	Mutation	Commonest alteration overall ("guardian of the genome")
RB1, PTEN	Mutation, SV, CN loss	Loss of ≥2 of TP53, RB1, PTEN = aggressive variant prostate cancer (AVPC) ; overlaps neuroendocrine prostate cancer (NEPC)
SPOP (± CHD1 loss)	Mutation	Commonest mutated suppressor in primary prostate cancer
BRCA1/2	Often germline first hit + somatic second	HR loss → deletions with flanking microhomology; synthetic lethality with PARP inhibitors — FDA-approved in metastatic prostate cancer
ATM	Loss	Included in PARP trials but little benefit — distinct from BRCA
CDK12	Somatic loss	Tandem duplications ; aggressive; poor response to hormones, chemotherapy and PARP inhibitors; immunotherapy under study
MMR genes	Deficiency → microsatellite instability	↑ mutational burden; FDA-approved indication for immunotherapy
AR	Amplification of gene and regulatory plexus (extrachromosomal circular DNA); mutation, SV	Late, under AR-targeted pressure; major driver of ADT resistance

- **TP53 is the commonest alteration across prostate cancer; SPOP is the commonest mutated suppressor in primary tumours** — the difference reflects disease stage.

- **Lineage plasticity** → **NEPC** (androgen-independent, highly aggressive) is a key resistance route; rarely present at diagnosis, usually **emerges on treatment** — **poor prognostic sign**.
- A "**long tail**" of rare alterations (<3% each) plus global expression and epigenomic changes complete the picture.

KEY POINTS: PROSTATE CANCER

- Driver alterations define genomic subtypes.
- mCRPC acquires AR amplification under AR-targeted pressure.
- NEPC/AVPC: loss of RB1, PTEN, TP53.
- BRCA1/2 → PARP inhibitors; MMR deficiency → immunotherapy.

Renal Cancer Somatic Alterations**RCC SUBTYPES: GENOMICS**

Subtype	Share	Genomic hallmark	Notes
Clear cell	75–80%	Near-universal 3p loss + mutation of remaining allele: VHL, PBRM1, BAP1, SETD2	Pseudohypoxia; belzutifan (HIF-2α inhibitor)
Papillary	15–20%	MET (7q) activation — especially former type 1; former type 2: CDKN2A silencing, SETD2, TFE3 fusions, NRF2 pathway, CpG island methylator phenotype with FH loss	MET-active TKIs effective; WHO has dropped the type 1/2 split (admixed)
Chromophobe	Rare	Mitochondrial DNA and mTOR pathway alterations; TP53, PTEN, TERT-promoter rearrangement	Distal nephron origin ; slow, treatment-resistant; mTOR-directed therapy rational
Translocation	Rare	Mit/TFE fusions ; 9p21.3 deletion; few other alterations	Young, female ; mimics other subtypes; NRF2 antioxidant response → resistance; no targeted therapy
Sarcomatoid / rhabdoid (any subtype)	—	↑ BAP1, CDKN2A deletion, MYC programmes	Aggressive but immune-inflamed (PD-L1+) → immunotherapy-responsive

CLEAR CELL RCC DRIVER GENES

Gene	Frequency	Prognosis / biology
VHL	Point mutation 60–70% + epigenetic silencing 5–10% (all RCC: ~60% inactivated)	Pseudohypoxia; germline VHL syndrome → >70% lifetime RCC risk
PBRM1	~40%	Better outcome ; ↑ angiogenesis genes, non-infiltrated tumours; mutually exclusive with BAP1
BAP1	~10%	Worse outcome
SETD2	~10%	Worse outcome

- PBRM1, BAP1, SETD2 are **chromatin modifiers** (repair, transcription), early events. Metastases carry more VHL, PBRM1, SETD2, BAP1, KDM5C, TP53, PTEN, TSC1, TET2, ARID1A mutations than primaries.
- **Transcriptomic subgroups** (angiogenesis, T-effector/IFN-γ immune, myeloid inflammation) predicted PFS differentially across atezolizumab ± bevacizumab vs sunitinib (**IMmotion150**); **IMmotion151** multi-omics defined **seven molecular subsets** (angiogenesis, immune, cell cycle, metabolism, stromal); **JAVELIN Renal 101** found immunomodulatory and angiogenesis signatures — aim: select patients for VEGF vs checkpoint therapy.

Bladder Cancer Somatic Alterations

- **Urothelial carcinoma:** commonest urinary-tract malignancy — **90% bladder, 10% upper tract** (mainly renal pelvis, ureter).
- **TCGA 2014 (MIBC): high somatic mutation rate;** recurrent pathway hits in **receptor tyrosine kinase/ MAPK** (FGFR, ERBB) and **PI3K/AKT/mTOR**; four expression clusters — cluster I **papillary-like, FGFR3-enriched**; I and II ERBB2-enriched; cluster III **basal/squamous-like** (like basal breast cancer).
- **TCGA 2017:** commonest altered genes **TP53** (most frequent), KMT2D, KDM6A, ARID1A, PIK3CA, KMT2C, RB1, EP300, FGFR3, STAG2, ATM; mutational load driven by **APOBEC-signature mutagenesis**; five subtypes (basal squamous, luminal papillary, luminal-infiltrated, luminal, neuronal).
- **Consensus classification (six subtypes): basal/squamous (35%, largest),** luminal papillary (24%), luminal unstable (15%), stroma-rich (15%), luminal non-specified (8%), **neuroendocrine-like (3%)** — the basis for future trials.

TARGETABLE ALTERATIONS IN UROTHELIAL CARCINOMA

Target	Frequency	Significance
FGFR (FGFR1–4 receptor tyrosine kinases)	15–20% advanced bladder; up to 37% upper tract	Enriched in luminal papillary tumours with low T-cell infiltration and poor immunotherapy response ; erdafitinib (pan-FGFR TKI) active in FGFR3/FGFR2 mutation or FGFR3 fusion → test all advanced UC for FGFR
HER2 (ERBB2)	9–12% MIBC (mutation or amplification)	Overexpression → worse outcome; no approved HER2 agent in UC (trials ongoing)
DDR genes	34% (TCGA); BRCA1/2 ~5% upper and lower tract; ~10% germline DDR variants	ERCC2 (NER) mutation 10–18% → cisplatin sensitivity ; DDR alterations → better outcomes → bladder-sparing trials; PARP inhibitors mixed

Testicular Cancer Somatic Alterations

- **Commonest solid malignancy of young males;** seminoma vs non-seminoma.
- **Low mutation rate,** distinct methylation profiles, frequent uniparental disomy. Commonest changes: **gain of 12p**; **KIT, KRAS, NRAS** mutations (especially seminoma). **TP53 mutations are rare**, unlike other solid tumours.
- **miR-371a-3p: >90% sensitivity and specificity,** outperforms classical serum markers for diagnosis and relapse monitoring; validation ongoing.

KEY POINTS: RCC, BLADDER, TESTICULAR CANCER

- VHL is the commonest clear cell RCC mutation → pseudohypoxia.
- PBRM1, BAP1, SETD2 alter prognosis.
- Papillary RCC: MET pathway.
- Urothelial: FGFR, HER2, DDR targets.
- Testis: few mutations; miRNA predicts relapse.

Germline Alterations

HEREDITARY SYNDROMES RELEVANT TO GU CANCER

Syndrome	Gene	Inheritance	GU tumours / features
Hereditary breast and ovarian cancer (HBOC)	BRCA2 > BRCA1	—	Prostate (risk clearer for BRCA2; BRCA1 controversial); also pancreas, melanoma
Lynch syndrome	MMR genes	—	Prostate and urothelial cancer, colorectal

Syndrome	Gene	Inheritance	GU tumours / features
Hereditary prostate cancer	HOXB13	—	Prostate; carriers → consider earlier PSA screening
Von Hippel-Lindau	VHL (suppressor)	AD	Clear cell RCC + other tumours; belzutifan approved
Hereditary papillary renal carcinoma (HPRC)	MET (activating)	AD	Papillary type 1 RCC
Hereditary leiomyomatosis and RCC (HLRCC)	FH (suppressor)	AD	Leiomyomas, leiomyosarcomas; papillary type 2 RCC , collecting-duct carcinoma
Birt-Hogg-Dubé	FLCN (folliculin)	AD	Genodermatosis; hybrid oncocytic , chromophobe, clear cell RCC, oncocytoma
Hyperparathyroidism–jaw tumour	HRPT2	AD	Parathyroid adenoma, jaw fibro-osseous tumours; cysts, hamartomas, mesoblastic nephroma, late Wilms', papillary RCC
Constitutional chromosome 3 translocation	Unknown (? VHL)	—	Clear cell RCC
Familial papillary thyroid cancer	Unknown	—	Papillary RCC, oncocytoma

- Prostate **genome-wide association study (GWAS)** loci are numerous but not yet clinically used.
- **Urothelial heritability is unclear** (shared environment confounds); Lynch is the known exception.
- **Testicular germ cell tumours are highly heritable** — via **many low-risk loci**, not a single syndrome; **Klinefelter association is controversial**.

Concept — Across GU tumours, the **germline syndrome names the somatic driver** (VHL → *clear cell*, MET → *papillary*, FH → *type 2 papillary*, BRCA → *prostate*), and the **driver's pathway names the drug** (HIF-2α, MET-TKI, PARP inhibitor, FGFR inhibitor, immunotherapy for MMR loss).

GENERAL PRINCIPLES OF CLINICAL NGS ASSAYS

- Three classes: **DNA panels**, **RNA panels**, **cell-free DNA (cfDNA) panels**.

DNA-Sequencing Panels

DNA ASSAY BREADTH (FIG. 25.13)

Assay	Coverage	Use
Whole-genome sequencing (WGS)	Entire genome	Most comprehensive (complex SVs, CN); most expensive — research only
WES	Exons	Cost ↓ by ~2 orders of magnitude; basis of early cancer studies
Targeted panels	Selected genes/exons (amplicon- or capture-based enrichment)	Most commercial panels
Germline panels	Small gene sets or hotspots (e.g. BRCA1/2)	Personal/family history; rarely change immediate treatment (exceptions: BRCA, radiotherapy contraindications); inform surveillance; genetic counselling recommended

- **Tumour-only sequencing is common**: many commercial panels **omit matched germline**, so somatic and germline variants cannot be separated — often not highlighted in reports.

Interpretation of Clinical DNA-Sequencing Results

- **Variant allele frequency (VAF)** = proportion of reads carrying the altered allele; shaped by **zygosity**, **CN** (amplification), **loss of heterozygosity (LOH)** and **tumour purity**.

- **Worked example:** heterozygous, CN-neutral alteration (50% of tumour DNA) × 60% purity → **VAF 0.5 × 0.6 = 30%**.

READING A VAF

VAF	Interpretation
High	Clonal — arose early, carried by most daughter cells
Low	Subclonal — later event
~50% (no germline data)	Caution — expected for a heterozygous germline variant
~100%	Probably homozygous germline
Single-digit % in cfDNA	Consider clonal haematopoiesis rather than tumour

- **Variants of unknown significance (VUS)** are common: a **high-VAF VUS in a gene frequently altered in that cancer is more likely pathogenic** (selection pressure). **Molecular tumour boards** help.

RNA-Sequencing Panels

- **Whole-transcriptome sequencing** (comprehensive, costly — research discovery) → **targeted panels** (cheaper, deeper, more accurate).
- Expression carries information DNA cannot; **cell-cycle/proliferation signatures underpin most clinical prognostic biomarkers in localised prostate cancer**. Older assays use arrays or quantitative polymerase chain reaction (qPCR) rather than RNA-seq.

Cell-Free and Circulating Tumour DNA

- Metastatic biopsies are infrequent → **cfDNA "liquid biopsy"** (blood; increasingly urine for GU cancer), mostly targeted panels.
- **Only a small (sometimes tiny) fraction of cfDNA is circulating tumour DNA (ctDNA)** → very deep sequencing; **more false positives and negatives** than tissue, worse without germline.
- **Clonal haematopoiesis (CHIP)**: common age-related myeloid mutations; since most non-tumour cfDNA comes from leucocytes, they **mimic somatic tumour variants** at single-digit VAF.
- **ctDNA fraction is prognostic** across cancers → **minimal residual disease (MRD)** assays: **WES of primary → bespoke hypertargeted panel of clonal alterations → ultra-deep sequencing after definitive treatment** → guide adjuvant therapy.
- Emerging: **cfDNA methylation** (confounded by ctDNA fraction) and **fragmentomics** (fragment size and position; detects cancer even at low ctDNA fraction).

Circulating Tumour Cells

- Haematogenous spread requires tumour cells in blood. Detection by **enrichment** (e.g. anti-EpCAM antibody, size) or **capture-free imaging**; **enumeration is prognostic**.
- Molecular profiling: **AR-V7** (splice variant lacking ligand-binding domain; constitutively active), neuroendocrine markers, **PD-L1 on RCC circulating tumour cells (CTCs)**.

CLINICAL APPLICATION OF MOLECULAR BIOMARKERS IN GENITOURINARY CANCERS

DIAGNOSTIC VS PROGNOSTIC VS PREDICTIVE BIOMARKERS

	Diagnostic Biomarkers	Prognostic Biomarkers	Predictive Biomarkers
Question	Is cancer present?	How aggressive?	Will this treatment help?
Metrics	Sensitivity, specificity, PPV, NPV	Recurrence-/progression-free and overall survival	Biomarker–treatment interaction

	Diagnostic Biomarkers	Prognostic Biomarkers	Predictive Biomarkers
Prostate examples	PSA (main; limited sensitivity, especially specificity); urine TMPRSS2-ERG, PCA3	Preoperative PSA, T stage, Gleason → NCCN risk groups ; molecular and artificial-intelligence (AI) tests	BRCA1/2 → PARP inhibitor; PORTOS (postoperative radiotherapy dose escalation, validated in SAKK 08/09); AI pathology → short-term ADT; germline HSD3B1

- **Diagnosis of clinically significant (intermediate/high-risk) disease is what matters**; better detection of low-risk cancer worsens overdiagnosis. With **multiparametric MRI (mpMRI) and fusion biopsy**, the role of urine tests is less clear.
- **Prognostic**: treatment usually intensifies as prognosis worsens, or is de-intensified/omitted (active surveillance) — but this is not universal, hence predictive markers.
- **A biomarker can be both**: NCCN risk groups predict ADT duration with radiotherapy — **short-term ADT only for unfavourable intermediate risk, long-term only for high risk**.
- **Relative vs absolute risk reduction**: with a fixed relative risk reduction (RRR), absolute benefit scales with baseline risk. **30% metastasis risk × 50% RRR = 15% absolute risk reduction (ARR) — measurable; 3% × 50% = 1.5% — undetectable** in most trials.
- **HSD3B1 (1245A>C)** germline variant boosts dihydrotestosterone synthesis → **ADT resistance**, yet **greater sensitivity to abiraterone**.
- **Statistical proof** of prediction needs a **biomarker-treatment interaction** test in a randomised trial including both biomarker groups — large samples, so many phase III trials enrol **biomarker-positive patients only**.

Fig. 25.14 Predictive and Prognostic Biomarkers

- Interaction plots of 10-year recurrence-free survival against biomarker score: **parallel treated/untreated lines = prognostic only** (with or without overall benefit); **intersecting lines = predictive** — benefit in high-score, none in low-score patients.

Concept — A prognostic marker tells you the **baseline risk**; a predictive marker tells you the **treatment effect**. Many "predictive" uses of prognostic markers work only through absolute risk — the same relative benefit is invisible when baseline risk is low.

KEY POINTS: BIOMARKERS

- Liquid biopsies allow serial, non-invasive monitoring with important limits.
- Interpret panel results by pathogenicity evidence, VAF and presence of germline sequencing.
- Prognostic → risk; predictive → treatment benefit; some do both.